Cancer variant analysis

COURSE INTRODUCTION

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Course etiquette



Video on when possible



Mute when not speaking

Use your **first** and **last** name as your Zoom ID
Right-click on yourself > Rename..

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Code of conduct

- Learning can only be done in a **safe environment**
- We abide to the ELIXIR **code of conduct** during the course
- If you witness unacceptable behaviour that can't be dealt with with the people involved in this course, contact the COC oversight group
- More info: here



ELIXIR Code of Conduct Principles

Trainers/organizers/Helpers

- **Gregoire Rossier:** Training coordinator at SIB
- **Diana Marek:** Training coordinator at SIB
- **Geert Van Geest:** Training Project Manager at SIB
- **Flavio Lombardo:** Computational Biologist, Consultant
- **Deepak Tanwar:** Bioinformatics Specialist and Trainer at SIB



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Learning outcomes

- Understand the difference between **germline** and **somatic** variants and the implication of computational analysis
- **Perform** a somatic **variant analysis** on a **paired** sample (tumor – normal) with GATK4
- Perform a somatic **variant annotation** with VEP and use the results to filter possible **high-impact mutations** in the cancer genome

Learning experiences

- Lectures
- Polls + quiz questions
- Exercises



Communication

- Course website: https://sib-swiss.github.io/cancer-variants-training/
- Shared document
- Zulip

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Variant calling

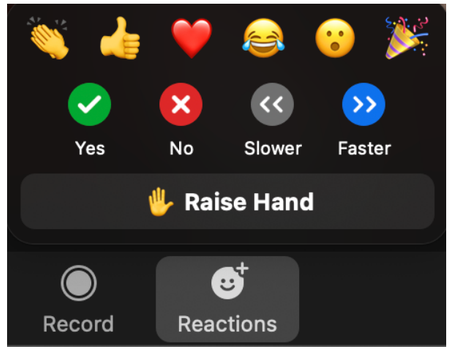
Short variant calling

Now that we have quality-controlled the BAM files, we can go ahead with the variant calling itself. For this, we used `mutect2` which is a somatic variant caller from GATK based on `HaplotypeCaller`. In addition to the expected bam files and references genome, the `mutect2` command requires some additional input:

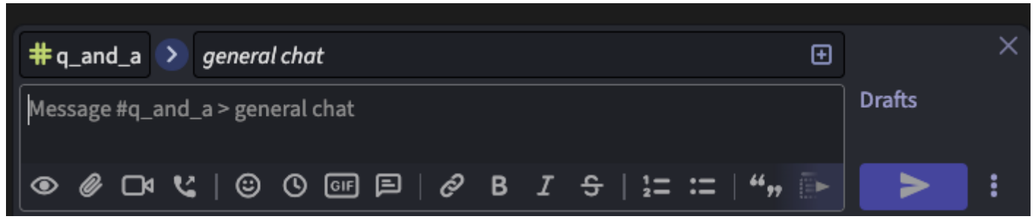
- `--intervals`: the intervals of our target regions. This is in the `interval_list` format.
- `-normal`: the sample name of the normal sample. So, the `SM` tag of the read group of the normal sample.
- `--germline-resource`: know sites of germline variants with their allele frequencies in the population. These are used to estimate the confidence of a germline variant in the normal sample.
- `--panel-of-normals`: A VCF generated from normal samples that contain sites with known technical artifacts. Ideally, you create a PON from your own normal samples, but this is typically recommended if you have more than 40 normal samples. Therefore, here, we use a pre-generated PON from the 1000 genomes project. More information on PON in this article.

Asking questions

- During lectures: zoom functionality – **but the chat will be ignored**



- During exercises: **#q_and_a channel on Zulip**
  - You can add a topic for your question, this will open a new conversation-topic.





Tell us about yourself:

- **Name** and affiliation
- **Research area** / current project
- **Experience** with variant analysis
- **What you hope to learn** from this course

…And consider the social channel in Zulip!